

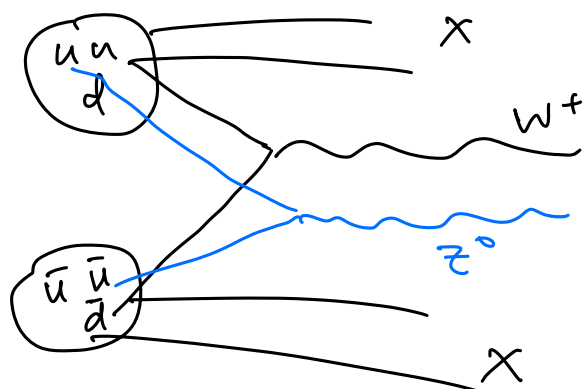
Discovery of W, Z bosons

SPPS @ CERN

$$\sqrt{s} = 540 \text{ GeV}$$

$$E_{beam} = 270 \text{ GeV}$$

$$\sqrt{s_{u\bar{u}}} \approx 170 \text{ GeV} \approx \frac{1}{3} \sqrt{s_{pp}} \text{ (in average)}$$



Drell-Yan process

you don't know
the x_u, x_d event-by-event.
 $\Rightarrow$ Lorentz boost unknown

$$P \longrightarrow \longleftarrow \bar{P} \longrightarrow Z$$

$$P_u = x_u P_P \quad P_{\bar{d}} = x_d P_{\bar{P}}$$

In general center-of-mass frame $\neq$ LAB

Lorentz boost always along Z axis

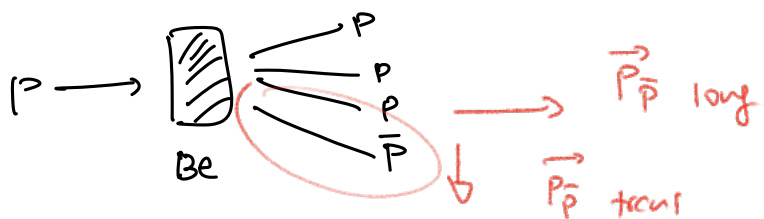
$\Rightarrow$ transverse momentum $\perp$ boost
will be invariant

Initial state $\vec{P}_P^T + \vec{P}_{\bar{P}}^T = \vec{0}$

Final state $\sum_i \vec{P}_{T,i} = \vec{0}$

Anti-proton production

$$P + P \rightarrow P + P + P + \bar{P}$$



1984 Nobel

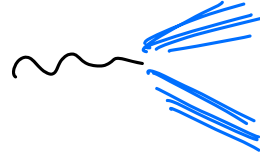
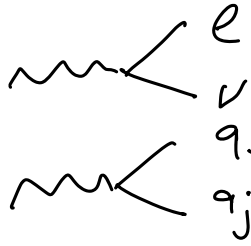
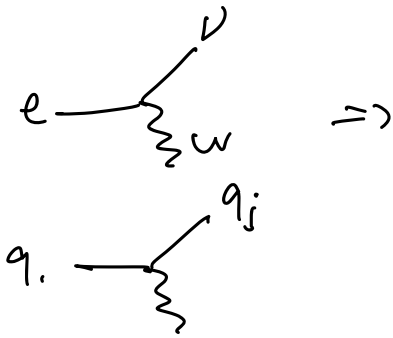
Rubbia - van Der Meer

Simon van der Meer

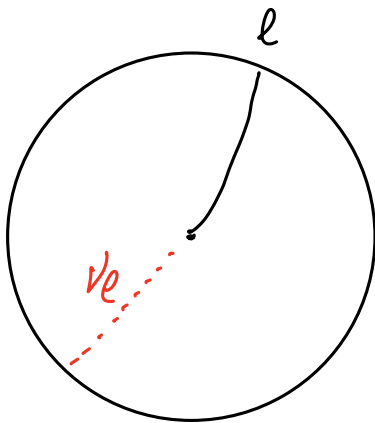
stochastic cooling
of anti-protons

$\Rightarrow$ key for $P-\bar{P}$
collisions at CERN.

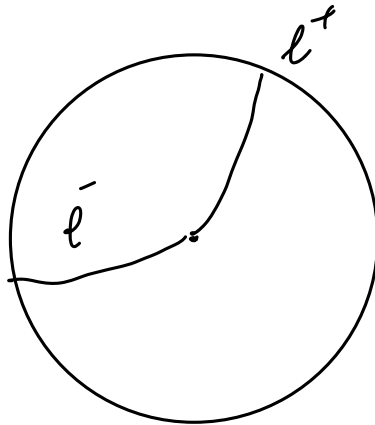
$p+p \rightarrow W/Z + X$
 $\hookrightarrow e^+e^-, \nu_e\bar{\nu}_e, q_i\bar{q}_i$
 $\hookrightarrow \bar{e}\nu, e^+\nu, q_i\bar{q}_j$



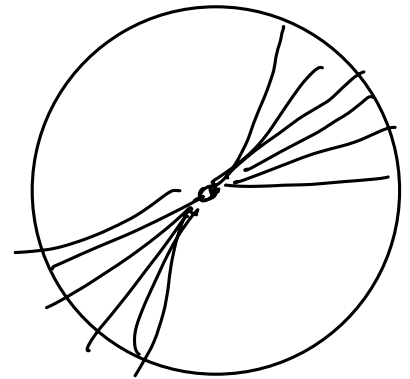
hadronic jets.



$W \rightarrow l\nu$

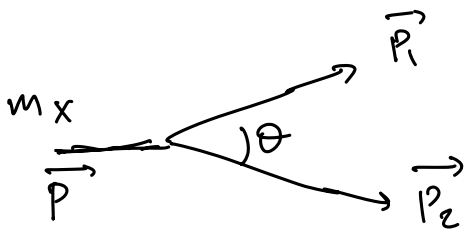


$Z^0 \rightarrow l\bar{l}$



$Z^0 \rightarrow \text{jets}$
 $W \rightarrow \text{jets}$

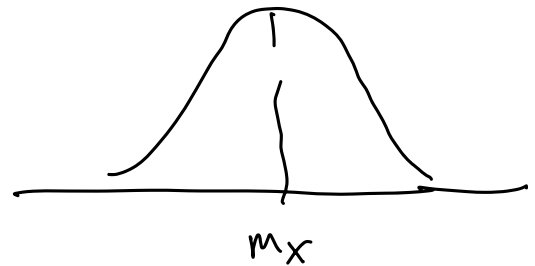
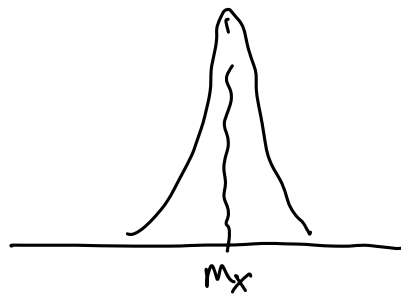
invariant mass
 e^+e^-



$$M_{12} = \sqrt{(\underline{p}_1 + \underline{p}_2)^2}$$

$$= \sqrt{2E_1E_2(1 - \cos\theta)}$$

No invariant mass
for $W \rightarrow l\nu$



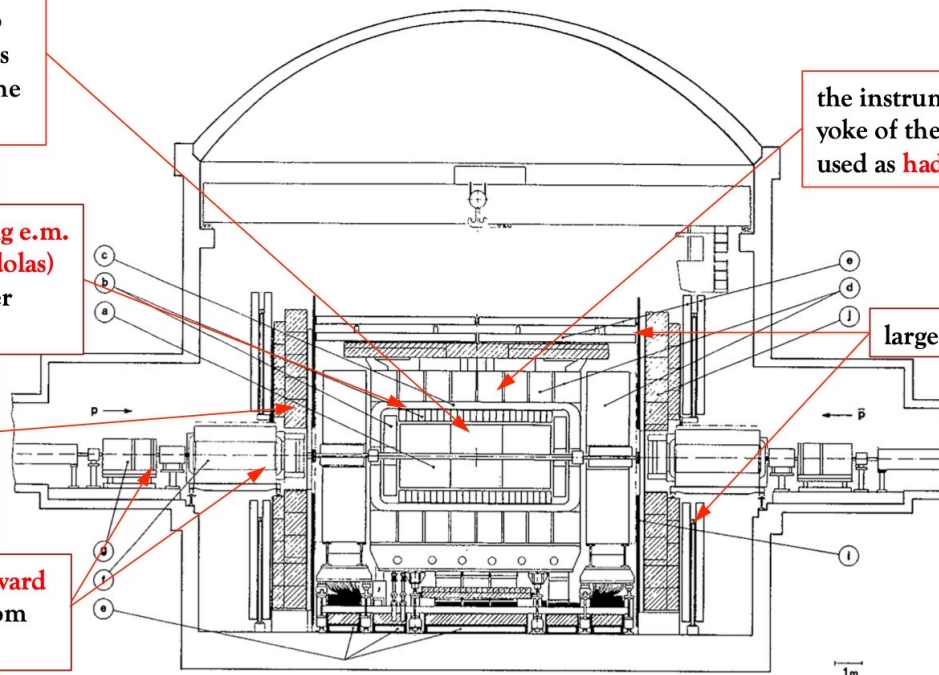


the **Central Detector**,
a big drift chamber to
track charged particles
in the 0.7 T field of the
dipole magnet

the fine grain **sampling e.m.
calorimeter (the Gondolas)**
readout by wave shifter
bars (BBQ)

end cap detectors
(the **bouchons**)

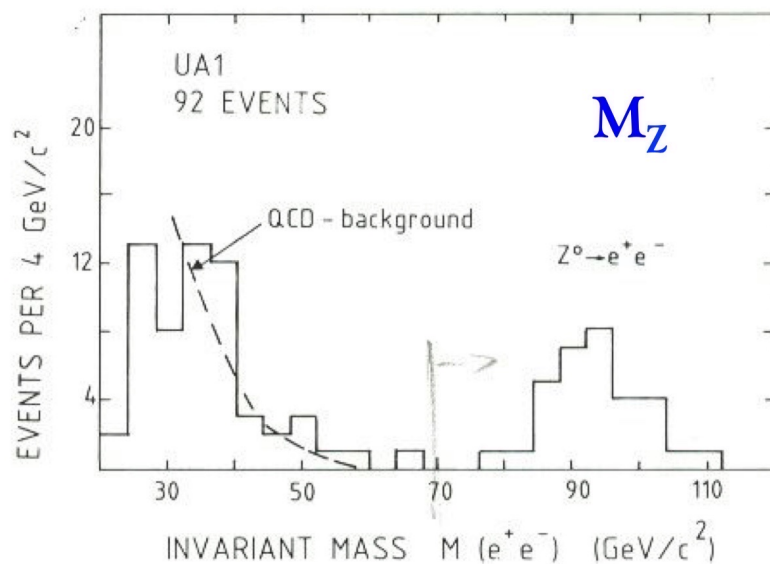
**forward and very forward
detectors** (up to 1° from
the beams)



the instrumented return
yoke of the dipole magnet
used as **hadron calorimeter**

large muon chambers

7

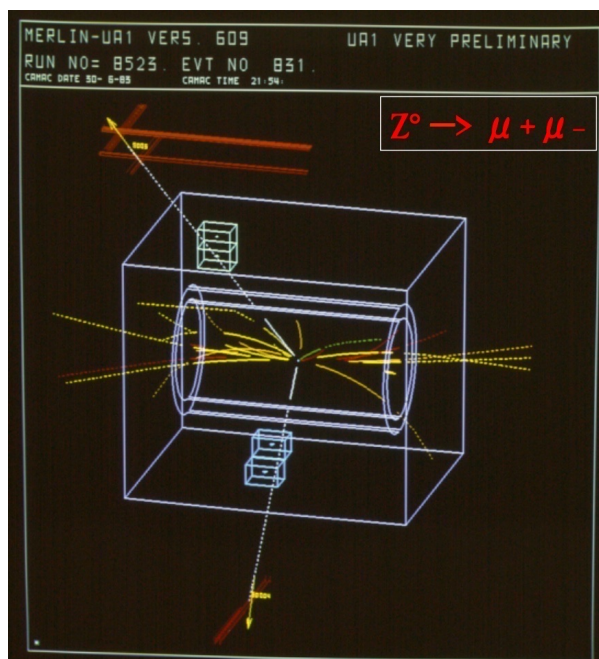
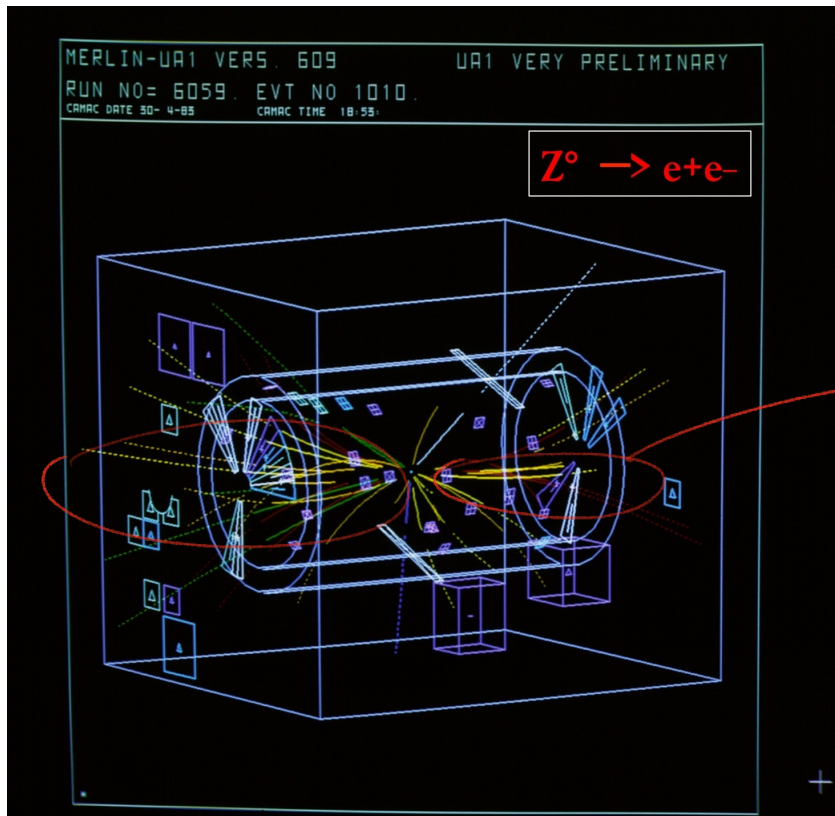


QCD background :

$$Z^0 \rightarrow q: \bar{q}; \quad (\text{jets}) \text{ hadrons}$$

experimental def of electron : charge track + EM calo.

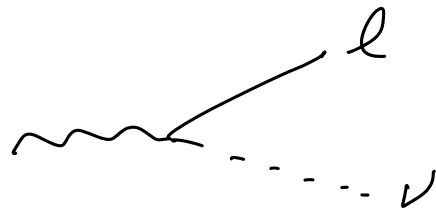
- real (core) electrons produced in hadron decays.
- fake electrons : pions exchanged for electron



W peak



W + jets



in W rest frame

$$e^- \leftarrow \overset{W^-}{\text{---}} \rightarrow \bar{\nu}_e$$

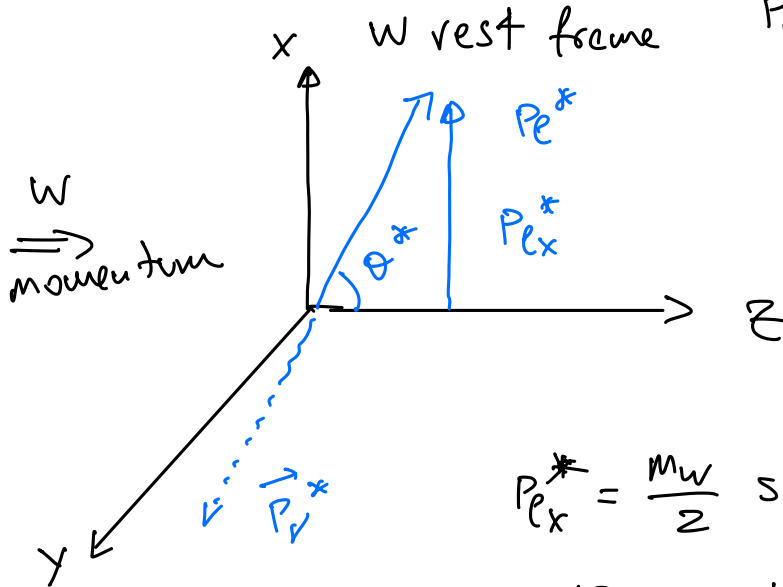
$$\vec{p}_e^* = -\vec{p}_\nu^*$$

$$E_\nu^* = E_e^* = \frac{m_W}{2}$$

$$E_\nu^* = p_\nu^*$$

$$E_e^* = p_e^*$$

$$p_{ey}^* = 0$$



$$p_{ex}^* = \frac{m_W}{2} \sin \theta^* \Rightarrow \sin \theta^* = \frac{p_{ex}^*}{m_W/2}$$

$$p_{ex}^{LAB} \equiv p_{ex}^* \quad \text{because } \perp \text{ boost.}$$

$$\frac{dn}{dp_{ex}^*} = \frac{dn}{d\theta^*} \frac{d\theta^*}{dp_{ex}^*}$$

$$\frac{dp_{ex}^*}{d\theta^*} = \frac{m_W}{2} \cos \theta^*$$

↳ distrib. of leptons
as function of θ^*
in W rest frame.

$$\cos \theta^* = \sqrt{1 - \sin^2 \theta^*}$$

$$= \sqrt{1 - \left(\frac{p_{ex}^*}{m_W/2}\right)^2}$$

$$= \frac{\sqrt{(m_W/2)^2 - (p_{ex}^*)^2}}{(m_W/2)}$$

$$p_{ex}^* \equiv p_{ex}$$

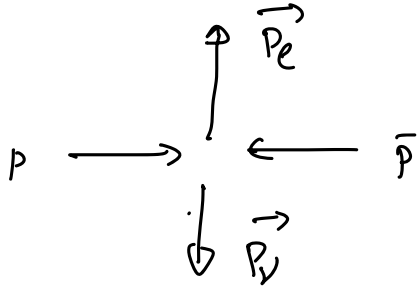
$$\frac{dn}{dp_{ex}} = \frac{dn}{d\theta^*} \frac{1}{\sqrt{(m_W/2)^2 - p_{ex}^2}}$$

leptons

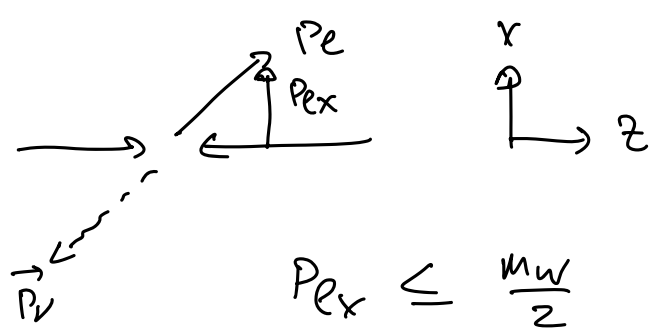
measure p_{ex} in LAB

Expect peak @ $p_{ex} = m_W/2$

p_{ex} : trans. mom of leptons.



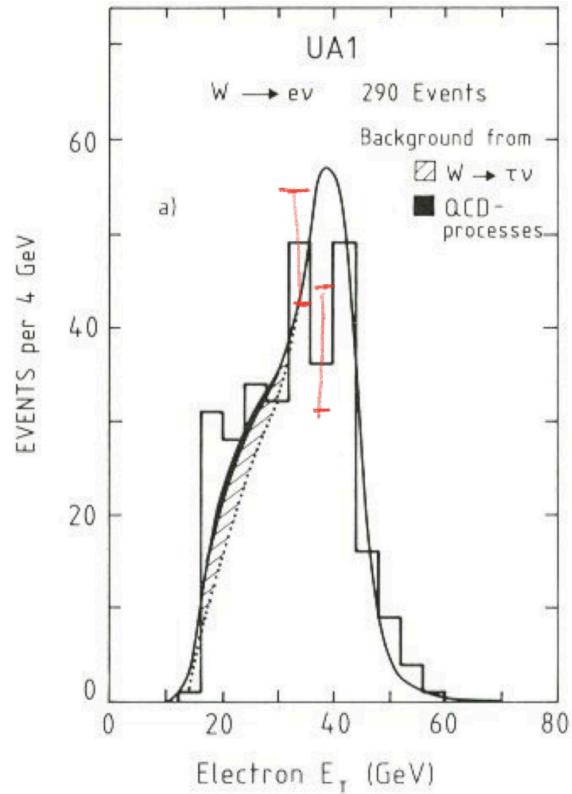
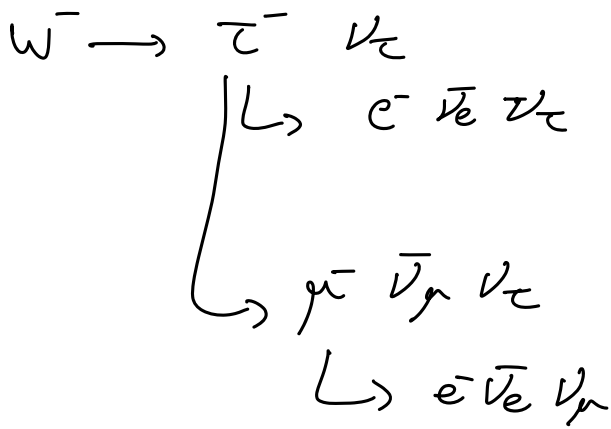
$$p_{ex} \equiv p_e = \frac{m_W}{2}$$



$$p_{ex} \leq \frac{m_W}{2}$$

QCD procceller:

fake electrons



Alternative approach for W

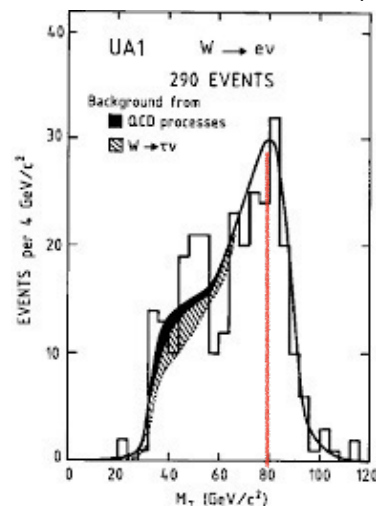
$$\sum_{\text{final state}} \vec{p}_{T,i} = 0 \Rightarrow \vec{p}_{T,\tau} + \sum_i \vec{p}_{T,i} = 0$$

$$\Rightarrow \vec{p}_{T,\tau} = - \sum_i \vec{p}_{T,i}$$

↳ all visible particles in final state.

transverse mass

$$m_W^T = \sqrt{(p_{eT} + p_{\nu T})^2} \leq m_W$$



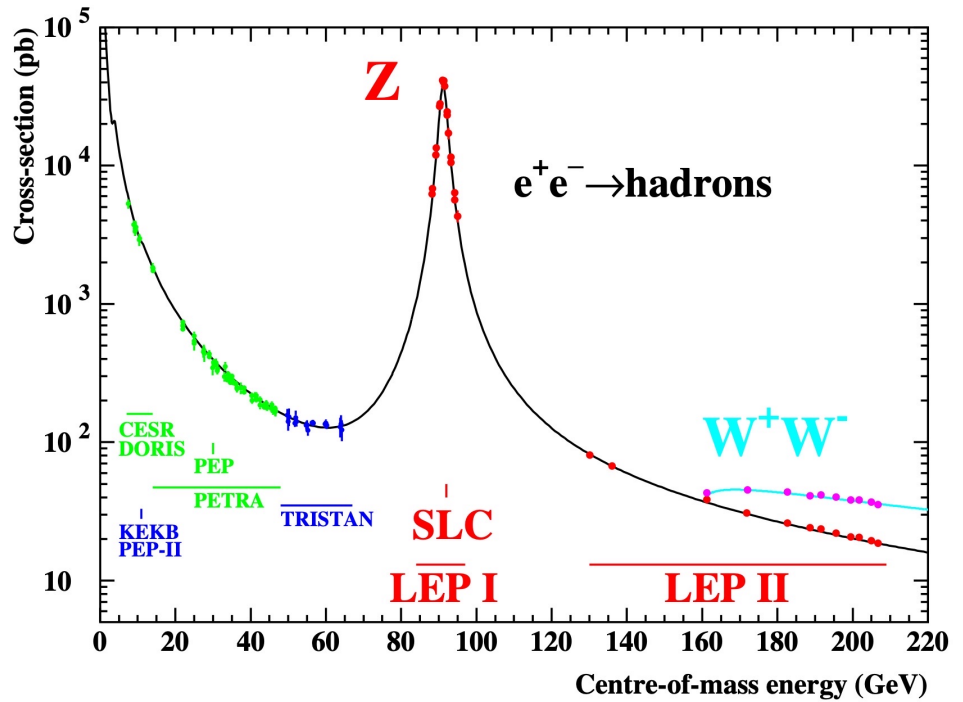
LEP machine @ CERN.

e^+e^-

$$\sqrt{s} = m_Z \approx 90 \text{ GeV}$$

1989 $\rightarrow$ 2002

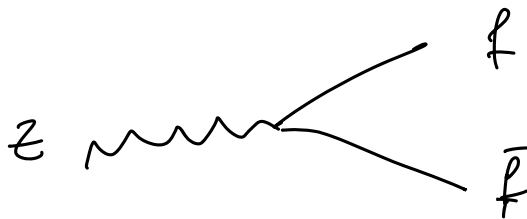
$$\sqrt{s} > m_Z \rightarrow 202 \text{ GeV}$$



Z DECAY MODES

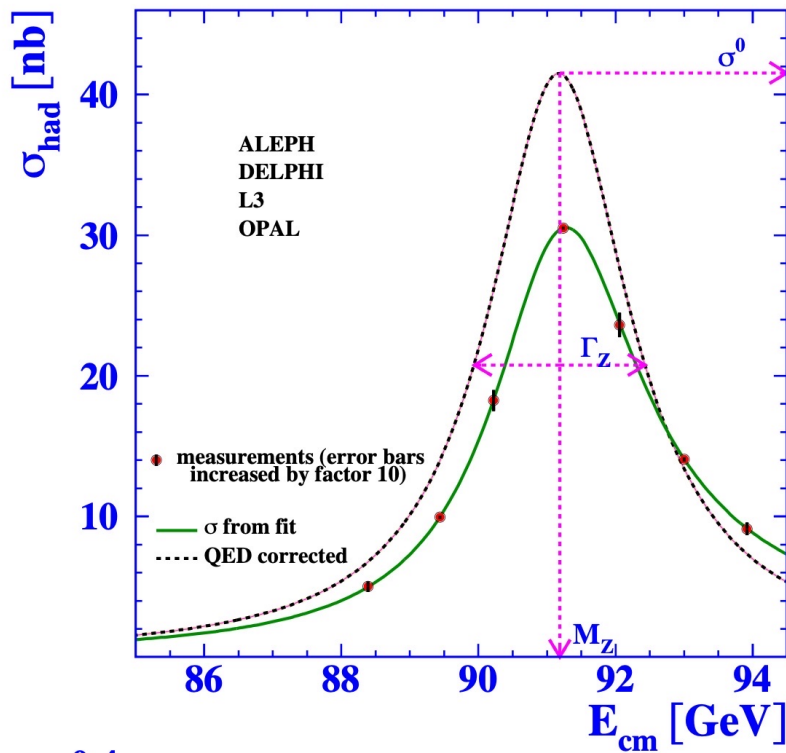
Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Γ_1 e^+e^-	[a] (3.3632 $\pm$ 0.0042) %	
Γ_2 $\mu^+\mu^-$	[a] (3.3662 $\pm$ 0.0066) %	
Γ_3 $\tau^+\tau^-$	[a] (3.3696 $\pm$ 0.0083) %	
Γ_4 $\ell^+\ell^-$	[a,b] (3.3658 $\pm$ 0.0023) %	
Γ_5 $\mu^+\mu^-\mu^+\mu^-$		
Γ_6 $\ell^+\ell^-\ell^+\ell^-$	[c] (4.55 $\pm$ 0.17) $\times 10^{-6}$	
Γ_7 invisible	[a] (20.000 $\pm$ 0.055) %	
Γ_8 hadrons	[a] (69.911 $\pm$ 0.056) %	

$$Z \rightarrow \nu\bar{\nu}$$



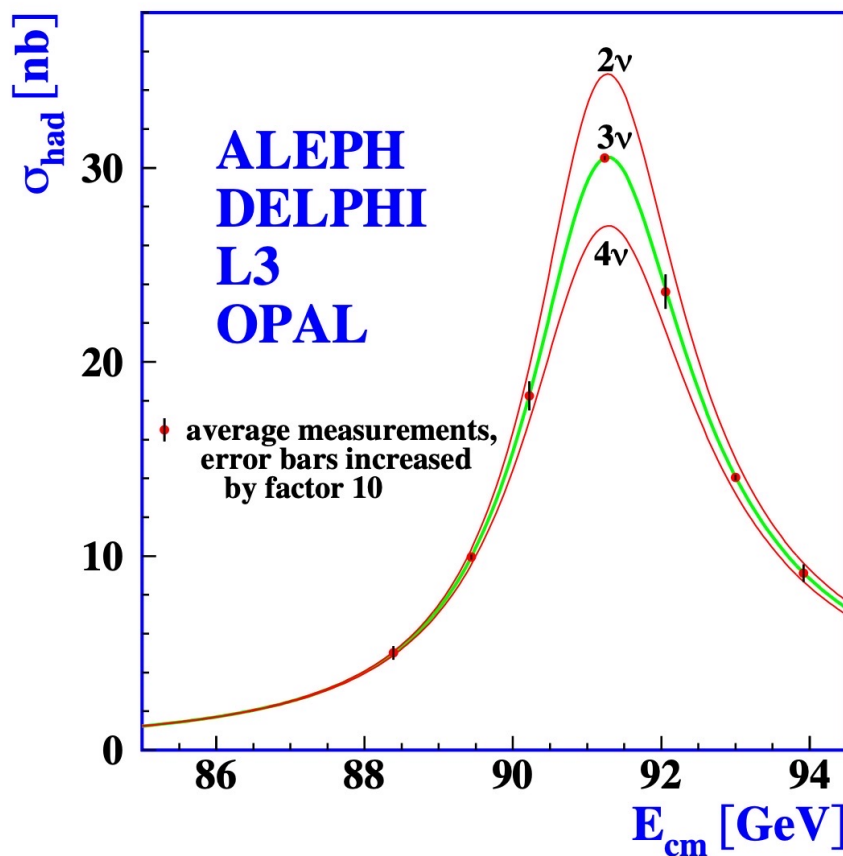
$$m_f \leq \frac{m_Z}{2}$$

you can probe new fermions with $m_f < m_Z/2$



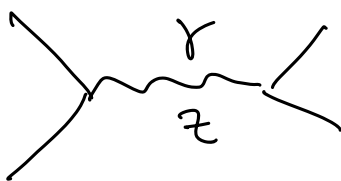
measure
total width Γ_Z

also compute
theoretically.



Fit to data
puts constraint
on $\Gamma(Z \rightarrow \nu\nu)$

$\Downarrow$
constraint on
 $\#$ neutrino
families.



$$\mu = \mu_\gamma + \mu_Z$$

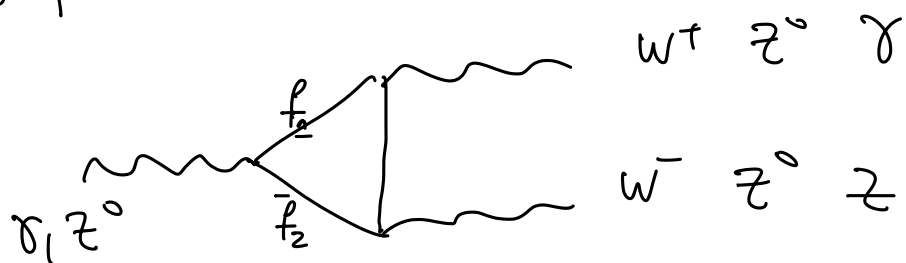
$$\# \text{ events} \propto |\mu_\gamma|^2 + |\mu_Z|^2 + \text{interference}$$

$$\begin{pmatrix} u \\ d \end{pmatrix} \quad \begin{pmatrix} c \\ s \end{pmatrix} \quad \begin{pmatrix} t \\ b \end{pmatrix} \quad \begin{pmatrix} q_7 \\ q_8 \end{pmatrix} \quad \# \nu \text{ fixed by lep} = 3$$

$$\begin{pmatrix} e^- \\ \nu_e \end{pmatrix} \quad \begin{pmatrix} \mu^- \\ \nu_\mu \end{pmatrix} \quad \begin{pmatrix} \tau^- \\ \nu_\tau \end{pmatrix} \quad ?$$

1st 2nd 3rd

In $SU(2)_L \times U(1)_Y$ SM each family has to be complete



anomalies disappear only if each family is complete